

such spots enlarge, coalesce and leaf blades especially from the tip towards base give a blasted appearance. Appear on the foliage. Rust pustules are oblong, brown, often formed in linear rows. They are also produced on the leaf sheaths, culms and stems.

Control Measures:

Blast: Spray Carbendazim @ 0.1% or Tricyclazole @ 0.05%.

Rust: Spray Mancozeb @ 0.2%. Warrant spray only if they appear at the early stages of crop growth.

Grain Smut (*Ustilago crameri*)

Symptom: Spikelets bigger in size, valvety gall like structure

Control Measures: Treating the seeds with Carboxin or Carbendazim @ 2g/kg

Downy Midew / Green ear (*Sclerospora graminicola*)

Symptom: Shredding of infected leaves, conversion of spikelets into leafy structure.

Control Measures: Treating the seeds with Ridomil MZ @ 6g/kg. Rouging out the affected plants.

Pests in Foxtail Millet

Army worm, cut worm and leaf scrapping beetle appear occasionally in serious form. In certain areas shoot fly occur, although it is not a regular pest.

Shoot fly

Symptom: The damage of the pest normally occurs in the seedling up to six weeks old crop. As a result of feeding, the central shoot starts drying and shows the typical appearance of 'dead heart' in early stage and profuse tillering in later stage, which are also affected. Damaged tillers may produce ear heads, but no grain.

Management: Early sowing within 7 to 10 days of the onset of monsoon. Another important practice is to increase the seed rate and destroy the 'deadheart' seedlings after removal, to maintain the optimum plant stand. Seed treatment with Imidacloprid @ 10-12 ml/kg of seed or Thiamethoxam 70 WS @ 3g/kg of seed may be used. Apply Carbofuran (Furadan 3G) or Phorate 10G at the time of sowing as soil application in the furrows @ 20 kg/ha.

Harvesting: Normally crop is ready for harvest in 80 - 90 days after sowing

Yield: Grain 20-25 q per ha under ideal condition & Straw 30-40 q per ha.

Compiled By:

K. Hariprasanna, Sangappa,
Shiwesh Kumar and Vilas A. Tonapi



हर कदम, हर डगर
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Foxtail Millet



ICAR - Indian Institute of Millets Research

Rajendranagar, Hyderabad - 500 030, Telangana.

Ph.: +91 - 040 - 2459 9300; Website:

www.millet.res.in

Foxtail Millet [*Setaria italica* (L.) Beauv.], one of the oldest of the cultivated millets in the world, is cultivated in about 23 countries in Asia, Africa and America. It is a self-pollinating, short duration, C₄ cereal, good as food for human consumption, feed for poultry and cage birds, and fodder for cattle. Foxtail millet ranks second among the millets production in the world and continues to have an important place in the world agriculture providing food to millions of people dependant on poor or marginal soils in southern Europe and in temperate, subtropical and tropical Asia. In India, it is grown mainly in Andhra Pradesh, Karnataka, Telangana, Rajasthan, Maharashtra, Tamil Nadu, Madhya Pradesh, Uttar Pradesh, and in a small extent in the northeast states of India. It is used as an energy source for pregnant and lactating women, and for sick people and children. Of late, the importance of foxtail millet is recognized as diabetic food. It is rich in dietary fibre, minerals, micronutrients, protein, and has low glycemic index (GI). Unlike rice, foxtail millet releases glucose steadily without affecting the metabolism of the body. The incidence of diabetes is found to be rare among the population consuming foxtail millet diet.

Local Names of Foxtail Millet

Language	Name
Hindi	Kangni, Kakum
Sanskrit	Kanguni
Kannada	Navane
Tamil	Tenai
Telugu	Korra, Korralu
Malayalam	Thina
Marathi	Kang, Rala
Gujarati	Kang
Bengali	Kaon
Oriya	Kanghu, Kangam, Kora
Punjabi	Kangni
Kashmiri	Shol

State Wise Popular Foxtail Millet Cultivars

State	Varieties popular in the State
Andhra Pradesh	SiA 3088, SiA 3156, SiA 3085, Lepakshi, SiA 326, Narasimharaya, Krishnadevaraya, PS 4
Bihar	RAU-1, SiA 3088, SiA 3156, SiA 3085, PS 4
Karnataka	DHFT-109-3, HMT 100-1, SiA 3088, SiA 3156, SiA 3085, PS 4, SiA 326, Narasimharaya
Rajasthan	Prathap Kangani 1 (SR 51), SR 1, SR 11, SR 16, SiA 3085, SiA 3156, PS 4
Tamil Nadu	TNAU 43, TNAU-186, Co (Te) 7, Co 1, Co 2, Co 4, Co 5, K2, K3, SiA 3088, SiA 3156, SiA 3085, PS 4
Telangana	SiA 3088, SiA 3156, SiA 3085, Lepakshi, SiA 326
Uttarakhand	PS 4, PRK 1, Sreelaxmi, SiA 326, SiA 3156, SiA 3085
Uttar Pradesh	PRK 1, PS 4, SiA 3085, SiA 3156, Sreelaxmi, Narasimharaya, S-114, SiA 326

Soil and Climate: Foxtail millet needs moderately fertile well drained soil for good yields, although it can grow on soils ranging from sandy to heavy clay soils. Grows better in place with annual rainfall of 500-700 mm. It cannot tolerant water logged condition or extreme drought. .

Time of Sowing:

Kharif: July-Aug. (Karnataka), July (TN, TS, AP),

2-3rd week of July (Maharashtra)

Rabi: August to September (Tamilnadu)

Seed Rate: 8-10 kg / ha for line sowing

15 kg / ha for Broadcasting

Seed Treatment: Ridomil @2g/kg, Carbendazim @ 2g / kg.

Spacing: 25-30 cm (row to row), 8 – 10 cm (plant to plant). The seed should be planted 2-3 cm in depth.

Manure and fertilizers: Apply Compost or farmyard manure @ 5-10 tonnes/ha about a month before sowing. Generally fertilizers recommended to get a good crop are 40 kg Nitrogen, 20 kg P O and 20 kg K O per ha. Soil test based fertilizers^{2 5} application is² recommended. Apply entire quantity of P O and half of Nitrogen at the time of sowing and^{2 5} remaining half of Nitrogen at 30 days after sowing.

Weeding and Intercultural operation: Two inter cultivations and one hand weeding in line sown crop is recommended. Intercultural operation using a tyne-harrow when crop is 30 days old is also recommended. In broadcast crop 1st weeding after 15 – 20 days of emergence of seedling and 2nd weeding 15-20 days after 1st weeding is recommended.

Irrigation: Kharif season crop require no or minimum irrigation. It is mostly grown as a rain-fed crop. However, if the dry spell prevails for longer period, then 1-2 irrigations to be given. Summer crop requires 2-5 irrigations depending upon soil type and climatic conditions.

Inter Cropping:

A.P/ Telangana: Foxtail Millet + Pigeon Pea (5:1)

Foxtail Millet + Groundnut (2:1)

Karnataka: Foxtail Millet + Cotton (5:1)

Foxtail Millet + Pigeon Pea (5:1)

Diseases

Blast (*Pyricularia setariae*), Rust (*Uromyces setariae*), Brown spot (*Cochliobolus setariae*).

Symptoms: Typical spindle shaped spots on leaf lamina. Under highly congenial conditions